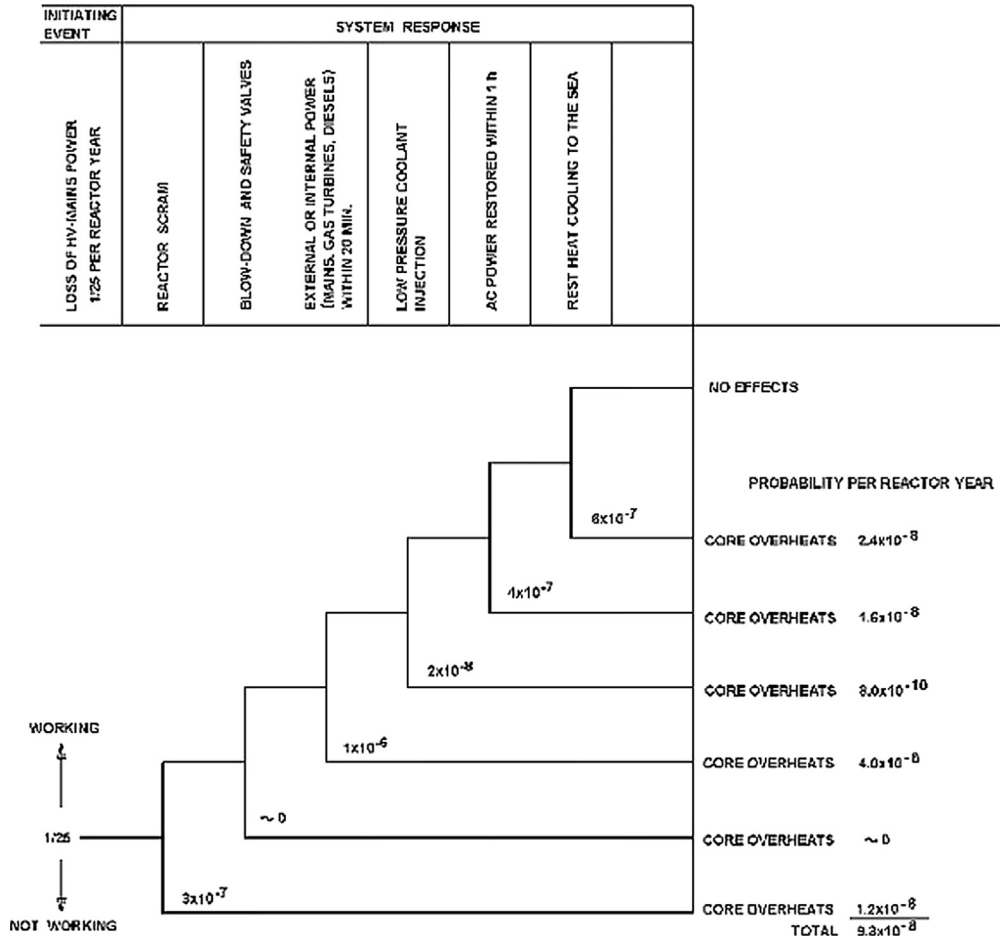


Figure 20.9 Example of an event tree for a hypothetical BWR.

20.6. EXERCISES

20.1. The gas space (volume 6000 m^3) in the PS system of a large BWR is filled with N_2 at 97 kPa and 25°C. Assume the same temperature in the dry and wet wells and that the heat of evaporation of water and its heat capacitivity are ~2.26 MJ/kg and ~4.18 kJ kg^{-1} $°C^{-1}$, respectively, up to ~100°C. The 2000 m^3 water in the condensation pool is at 25°C. The primary system contains steam at 287°C and 7 MPa (energy content ~2.8 MJ/kg steam). How many kg of such steam could be dumped into the condensation pool before the pressure in the PS system exceeds 0.1 MPa?

20.2. The amount of steam found in example 20.1 is released into an air-filled large dry containment where the pressure may not exceed 0.7 MPa. Assume steam and air in the containment can be treated as ideal gases. Estimate the necessary containment volume.

20.3. In the Chernobyl accident, reactor output increased from ~20% to ~100 times full power in about 4 seconds. Assume a constant reactivity excess during this time. a) What was the average reactor period? b) How much energy was released during this time?

20.4. KOH is normally used for pH control in VVER reactors. Which radioisotopes will be formed from potassium in the coolant?

20.5. ^{232}U is formed in a thermal ^{232}Th based breeder. a) By which chain of reactions is it formed? b) Will the choice between a continuous or discontinuous (once a year) reprocessing of the Th blanket affect the isotopic composition of the isolated uranium (assume that all other actinides remain in the blanket)?

20.6. Use eqn. (19.43) to estimate the maximum thermal efficiency for the Creys-Malville plant.

20.7. Assume that the probability of not restoring AC power within 20 minutes is reduced by a factor of 5 in Fig. 20.9. a) Which is now the risk dominating event chain? b) Is it better to reduce the total risk in Fig. 20.9 by this change or by improving the stability of the HV mains with a factor of 2 by e.g. building more power stations?

20.8. What is the fate of the gas released into the turbine condensers during regeneration of the charcoal absorbers in Fig. 20.6?

20.7. LITERATURE

J. K. Dawson and R. G. Sowden, *Chemical Aspects of Nuclear Reactors*, Butterworths, 1963.

S. Peterson and R. G. Wymer, *Chemistry in Reactor Technology*, Pergamon Press, Oxford, 1963.

Reactor Handbook, 2nd edn., Interscience, 1960?1964.

J. R. Lamarch, *Introduction to Nuclear Reactor Theory*, Addison-Wesley, 1966.

J. G. Wills, *Nuclear Power Plant Technology*, Wiley, 1967.

C. O. Smith, *Nuclear Reactor Materials*, Addison-Wesley, 1967.

G. I. Bell and S. Glasstone, *Nuclear Reactor Theory*, van Nostrand, 1970.

W. Fratzscher and H. Felke, *Einführung in die Kerntechnik*, VEB Deutscher Verlag für Grundstoffindustrie, Leipzig, 1971.

U.N. Conferences on peaceful uses of atomic energy, 1955, 1958, 1965, 1972.

J. R. Lamarch, *Introduction to Nuclear Engineering*, Addison-Wesley, 1975.

IAEA, *Management of Radioactive Wastes from the Nuclear Fuel Cycle*, Vols. 1 and 2, Vienna, 1976.

Jahrbuch der Atomwirtschaft 1977, Handelsblatt GmbH, Verlag für Wirtschaftsinformation, Düsseldorf, Frankfurt, 1977.

Int. Conf. on Nuclear Power and its Fuel Cycle, Salzburg 1977, IAEA, Vienna, 1977.

R. A. Rydin, *Nuclear Reactor Theory and Design*, PBS Publications, 1977.

H. Edelhäuser, H. Bonke, I. Gans, O. Huber, K. J. Vogt, and R. Wolter, *Radioactive Effluents and Present and Future Radiation Exposure to the Population from Nuclear Facilities in the Federal Republic of Germany*, in Int. Conf. on Nuclear Power and its Fuel Cycle, Salzburg 1977, IAEA, Vienna, 1977.